

WEEK 2 PROJECT REPORT

DATA MODELING, DAX & SALES DASHBOARD

Project Title: Superstore Sales Data Analysis using Power BI

Tool Used: Microsoft Power BI Desktop

Dataset: Superstore Sales Dataset

1. INTRODUCTION

The objective of Week 2 was to transform the cleaned Superstore dataset into an interactive **Power BI Sales Dashboard**. The project involved creating additional date columns, developing DAX measures, designing KPI cards and charts, and adding interactive slicers.

The dashboard helps analyze sales performance, profitability, orders, quantity, product categories, regions, and customer segments.

2. DATA COLUMN CREATION

After completing the data cleaning process, additional columns were created from the **Order Date** column to support time-based analysis.

The following columns were created:

Column	Purpose
Order Year	Analyze sales by year
Order Quarter	Analyze sales by quarter
Order Month	Analyze sales by month
Order Day	Identify the day of the month
Order Day of Week	Identify the weekday

The duplicated **Order Date** column was transformed using Power Query functions such as:

Order Year

Date.Year([Order Date])

Order Quarter

Date.QuarterOfYear([Order Date])

Order Month

Date.MonthName([Order Date])

Order Day

Date.Day([Order Date])

Order Day of Week

Date.DayOfWeekName([Order Date])

The screenshot displays a Power BI Desktop interface. The main area shows a data table with the following columns: Year, Order Year, Order Quarter, Order Month, Order Day, and Order Day of Week. The data represents the first 26 days of January 2023. The right sidebar contains the 'APPLIED STEPS' pane, which lists the following transformations: Source, Promoted Headers, Changed Type, Changed Type with Locale, Changed Type with Locale1, Trimmed Text, Reordered Columns, Inserted Day, Added Custom, Added Custom1, Added Custom2, Added Custom3, Added Custom4, Added Custom5, and Changed Type1.

	Year	Order Year	Order Quarter	Order Month	Order Day	Order Day of Week
1	2023	2023	1	January	3	Tuesday
2	2023	2023	1	January	4	Wednesday
3	2023	2023	1	January	4	Wednesday
4	2023	2023	1	January	4	Wednesday
5	2023	2023	1	January	5	Thursday
6	2023	2023	1	January	6	Friday
7	2023	2023	1	January	6	Friday
8	2023	2023	1	January	6	Friday
9	2023	2023	1	January	6	Friday
10	2023	2023	1	January	6	Friday
11	2023	2023	1	January	6	Friday
12	2023	2023	1	January	6	Friday
13	2023	2023	1	January	6	Friday
14	2023	2023	1	January	6	Friday
15	2023	2023	1	January	7	Saturday
16	2023	2023	1	January	7	Saturday
17	2023	2023	1	January	9	Monday
18	2023	2023	1	January	9	Monday
19	2023	2023	1	January	10	Tuesday
20	2023	2023	1	January	10	Tuesday
21	2023	2023	1	January	11	Wednesday
22	2023	2023	1	January	13	Friday
23	2023	2023	1	January	13	Friday
24	2023	2023	1	January	13	Friday
25	2023	2023	1	January	13	Friday
26	2023	2023	1	January	13	Friday

These columns make it easier to perform temporal analysis and apply date-based filters in the dashboard.

3. CREATION OF MEASURES TABLE

A dedicated table named **_Measures** was created to organize all DAX measures in one location.

This improves the structure and readability of the Power BI data model.

The **_Measures** table was used to store the following measures:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Average Sales
- Profit Margin %

4. DAX MEASURES CREATED

4.1 Total Sales

The Total Sales measure calculates the total revenue generated.

Total Sales = SUM('samplesuperstore'[Sales])

The screenshot displays the Power BI interface with three main panes: Filters, Visualizations, and Data.

- Filters Pane:** Contains a search bar and three filter sections: "Filters on this visual" (showing "Total Sales is (All)"), "Filters on this page", and "Filters on all pages". Each section has an "Add data fields here" button.
- Visualizations Pane:** Features a "Build visual" section with icons for various chart types. Below this is a grid of visualization options. At the bottom, there are sections for "Value" (set to "Total Sales"), "Categories", "Tooltips", and "Drill through" (with "Cross-report" set to "Off" and "Keep all filters" set to "On").
- Data Pane:** Includes a search bar and a list of data sources: "Measures" and "samplesuperstore".

4.2 Total Profit

The Total Profit measure calculates the total profit.

Total Profit = SUM('samplesuperstore'[Profit])

The screenshot displays the Qlik Sense interface with three main panels: Filters, Visualizations, and Data.

- Filters Panel:** Contains a search bar and three filter sections: "Filters on this visual" (showing "Total Profit is (All)"), "Filters on this page", and "Filters on all pages". Each section has an "Add data fields here" button.
- Visualizations Panel:** Features a "Build visual" section with a grid of visualization icons. Below this is a configuration area for the selected visualization, including:
 - Value:** A dropdown menu showing "Total Profit".
 - Categories:** A section with an "Add data fields here" button.
 - Tooltips:** A section with an "Add data fields here" button.
 - Drill through:** Includes a "Cross-report" toggle (set to Off) and a "Keep all filters" toggle (set to On). Below these is an "Add drill-through fields here" button.
- Data Panel:** Contains a search bar and a list of data sources: "Measures" and "samplesuperstore".

4.3 Total Orders

The Total Orders measure calculates the number of unique orders.

Total Orders = `DISTINCTCOUNT('samplesuperstore'[Order ID])`

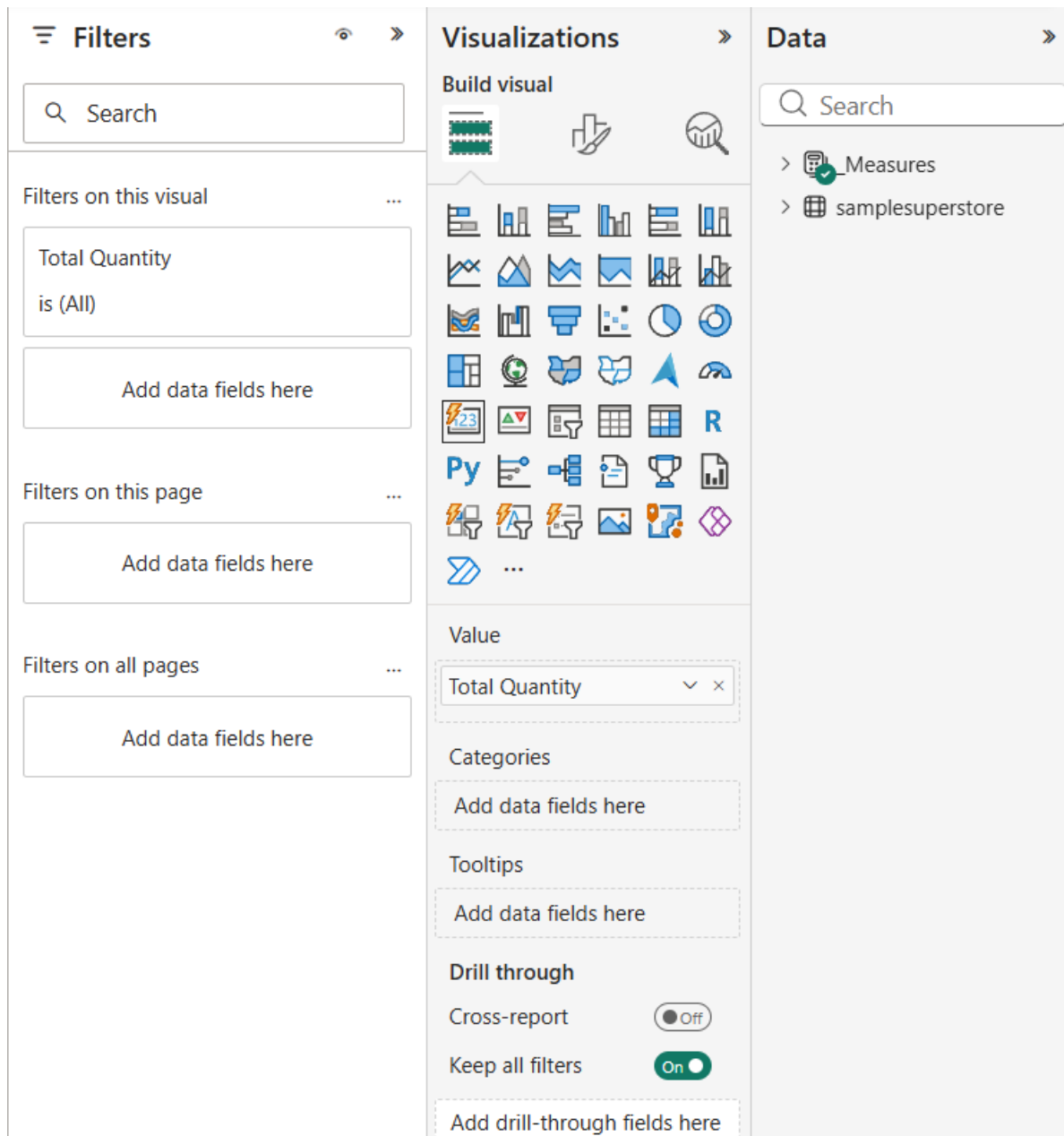
The screenshot displays the Power BI interface with three main panes: Filters, Visualizations, and Data.

- Filters Pane:** Contains a search bar and three filter sections:
 - Filters on this visual:** Shows 'Total Orders is (All)'.
 - Filters on this page:** Empty.
 - Filters on all pages:** Empty.
- Visualizations Pane:** Contains a 'Build visual' section with a grid of visualization icons. Below the grid are sections for 'Value' (set to 'Total Orders'), 'Categories', 'Tooltips', and 'Drill through' (with 'Cross-report' set to 'Off' and 'Keep all filters' set to 'On').
- Data Pane:** Contains a search bar and a list of data sources: 'Measures' and 'samplesuperstore'.

4.4 Total Quantity

The Total Quantity measure calculates the total quantity of products sold.

Total Quantity = SUM('samplesuperstore'[Quantity])



4.5 Average Sales

The Average Sales measure calculates the average sales value.

Average Sales = AVERAGE('samplesuperstore'[Sales])

4.6 Profit Margin %

The Profit Margin percentage measures profitability relative to total sales.

Profit Margin % = $\text{DIVIDE}([\text{Total Profit}], [\text{Total Sales}], 0)$

The measure was formatted as a **Percentage**.

The screenshot displays the Power BI interface with three main panes: Filters, Visualizations, and Data.

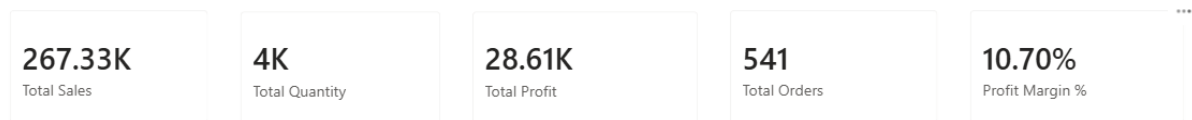
- Filters Pane:** Contains a search bar and three filter sections: "Filters on this visual" (with "Profit Margin %" and "is (All)"), "Filters on this page", and "Filters on all pages". Each section has an "Add data fields here" button.
- Visualizations Pane:** Features a "Build visual" section with a grid of visualization icons. Below this is a "Value" section where "Profit Margin %" is selected, and "Categories", "Tooltips", and "Drill through" sections, each with an "Add data fields here" button. The "Drill through" section includes toggle switches for "Cross-report" (Off) and "Keep all filters" (On).
- Data Pane:** Includes a search bar and a list of measures under the "Measures" category. The measures are: Average Sales, Profit Margin % (checked), Total Orders, Total Profit, Total Quantity, and Total Sales. Below the measures is a section for "samplesuperstore".

5. KPI CARDS

Five KPI Cards were created to provide a quick summary of business performance.

The KPIs included:

- 💰 **Total Sales**
- 📦 **Total Quantity**
- 💵 **Total Profit**
- 📦 **Total Orders**
- 📈 **Profit Margin %**



These KPI cards allow users to quickly understand the overall performance of the business.

6. VISUALIZATIONS CREATED

Several charts were created to analyze different aspects of the Superstore dataset.

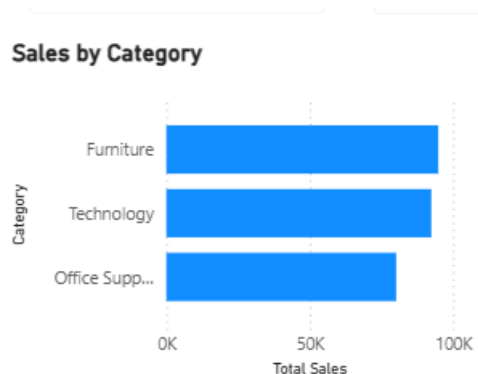
6.1 Sales by Category

Visual Type: Clustered Bar Chart

- **Y-axis:** Category
- **X-axis:** Total Sales

This chart compares sales performance across the major product categories:

- Technology
- Furniture
- Office Supplies



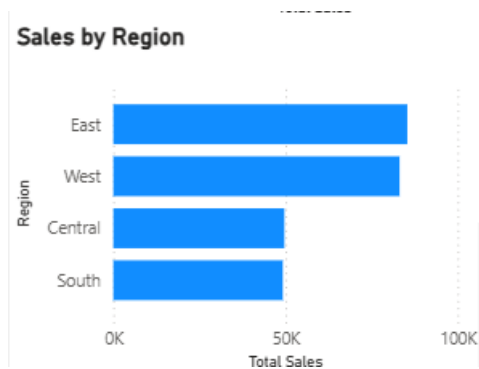
6.2 Sales by Region

Visual Type: Clustered Bar Chart

- **Y-axis:** Region
- **X-axis:** Total Sales

The chart analyzes sales across the four regions:

- Central
- East
- South
- West

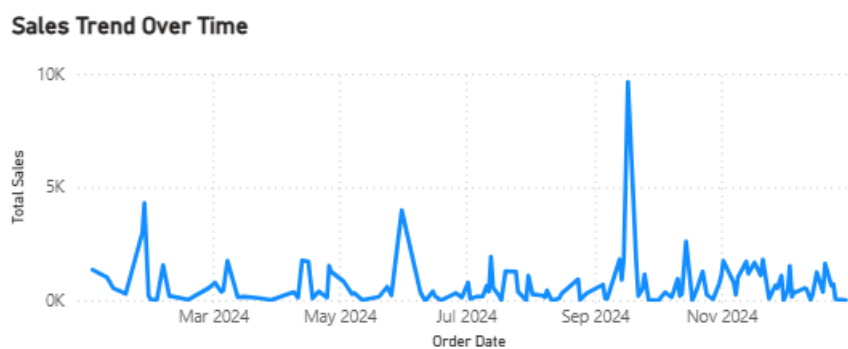


6.3 Sales Trend Over Time

Visual Type: Line Chart

- **X-axis:** Order Date
- **Y-axis:** Total Sales

This visualization shows how sales changed over time and helps identify trends and fluctuations.



6.4 Profit by Sub-Category

Visual Type: Clustered Bar Chart

- **Y-axis:** Sub-Category
- **X-axis:** Total Profit

This chart helps identify the most profitable product sub-categories.

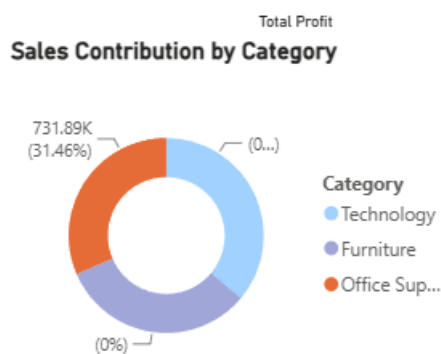


6.5 Sales Contribution by Category

Visual Type: Donut Chart

- **Legend:** Category
- **Values:** Total Sales

The donut chart shows the percentage contribution of each product category to total sales.



7. INTERACTIVE SLICERS

Interactive slicers were added to allow users to filter the dashboard dynamically.

The following slicers were created:

- **Region**
- **Category**

- **Segment**
- **Order Year**

The Order Year slicer was displayed as a **List**, allowing users to select individual years easily.

These slicers automatically update the KPI cards and visualizations based on user selections.

Region ☐ Central ☐ East ☐ South ☐ West	Order Year ☐ 2023 ☐ 2024 ☐ 2025 ☐ 2026	
Category ☐ Furniture ☐ Office Supplies ☐ Technology	Segment ☐ Consumer ☐ Corporate ☐ Home Office	

8. DASHBOARD DESIGN

A dashboard titled:

SUPERSTORE SALES DASHBOARD

Sales Performance Analysis

was created. The dashboard was organized into the following sections:

Top Section

- Dashboard Title
- KPI Cards

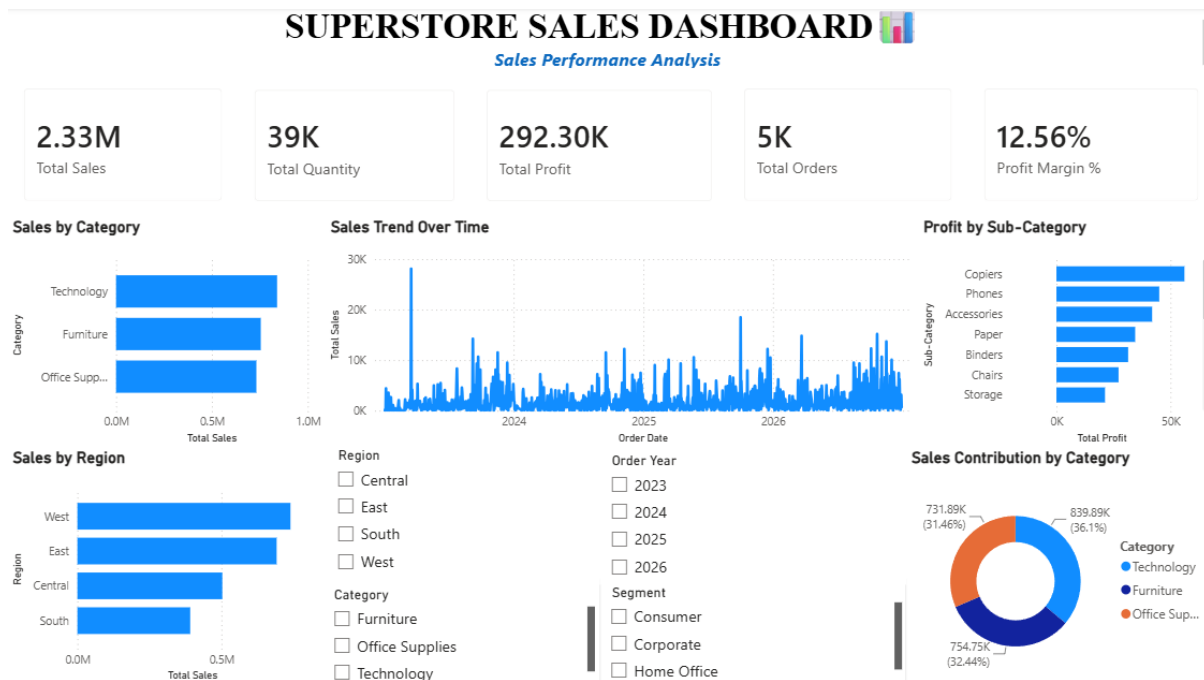
Analysis Section

- Sales by Category
- Sales Trend Over Time
- Profit by Sub-Category
- Sales by Region
- Sales Contribution by Category

Filter Section

- Segment
- Region
- Category
- Order Year

The visuals were arranged to create a clean and user-friendly dashboard.



9. KEY RESULTS

The overall dashboard showed the following business performance indicators:

KPI	Result
Total Sales	2.33M
Total Quantity	39K
Total Profit	292.30K
Total Orders	5K
Profit Margin	12.56%

10. CONCLUSION

In Week 2, the cleaned Superstore dataset was successfully transformed into an interactive **Power BI Sales Dashboard**. DAX measures were created to calculate important business metrics such as Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin. KPI cards and multiple visualizations were then developed to analyze sales and profitability. Interactive slicers were added to enable dynamic filtering and exploration of the data.

Overall, the project demonstrates the use of **Power BI for business intelligence, DAX calculations, data analysis, and interactive dashboard development**.